

MOH IRSHAD

M.Tech (Fuel & Energy Engineering), IIT (ISM) Dhanbad 📞 +91-77349-28875
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Professional Summary

M.Tech student in **Fuel & Energy Engineering** at IIT (ISM) Dhanbad with experience in process modelling, energy analysis, biomass conversion, and thermochemical processes. Currently using **Aspen Plus** to model and analyse biomass gasification, water-gas shift (WGS), and biogas reforming pathways. Academic work covers fuel technology, energy conversion, process simulation, mass and energy balances, energy efficiency, and exergy analysis. Prior industry experience includes production operations, plant data analysis, equipment performance monitoring, technical reporting, and day-to-day engineering activities. Seeking opportunities in **Fuel & Energy Engineering, Process Engineering, Energy Systems, Renewable Energy, and Clean Energy** applications.

Education

Indian Institute of Technology (Indian School of Mines), Dhanbad <i>M.Tech in Fuel & Energy Engineering</i>	2025 – 2027 (Expected) <i>Dhanbad, Jharkhand</i>
S.S. College of Engineering, Udaipur (RTU, Kota) <i>B.Tech in Mining Engineering</i>	2020 – 2024 <i>Udaipur, Rajasthan</i>
• CGPA: 9.41/10	

Relevant Coursework

- **Fuel & Energy:** Fuel Technology, Biofuel, Alternate Energy Systems, Power Plant Engineering, Thermochemical Conversion
- **Process & Energy Analysis:** Process Simulation, Mass & Energy Balances, Energy Conversion, Energy Efficiency, Energy Analysis
- **Renewable & Sustainable Energy:** Biomass Energy, Renewable Energy Systems, Sustainable Energy, Energy Transition

Experience

Indian Institute of Technology (Indian School of Mines), Dhanbad <i>M.Tech Researcher – Fuel & Energy Systems</i>	Aug 2025 – Present <i>Dhanbad, Jharkhand</i>
• Developing Aspen Plus process models to compare hydrogen-production pathways based on biomass gasification, water-gas shift (WGS), and biogas reforming. • Setting up feed specifications, thermodynamic property methods, process boundaries, and operating conditions to establish a common basis for cross-pathway process comparison. • Evaluating hydrogen yield, syngas composition, H_2/CO ratio, mass and energy balances, energy efficiency, and exergy efficiency through sensitivity analysis of key operating conditions.	
CSIR – National Chemical Laboratory (NCL), Pune <i>Research Intern – Conversion of Biorenewables to Value-added Chemicals</i>	Jun 2026 – Jun 2026 <i>Pune, Maharashtra</i>
• Completed a research internship on the conversion of biorenewable feedstocks into value-added chemicals under the supervision of Dr Satyam Naidu Vasireddy, Scientist-D, CSIR-NCL. • Studied biomass-derived feedstocks, catalytic conversion, reaction pathways, and experimental research methods relevant to value-added chemical production.	
Lakhanpur Masonry Stone Mine (M/s Shri Ram Stone Crusher) <i>Graduate Engineer Trainee</i>	Jun 2024 – Jun 2025 <i>Bharatpur, Rajasthan</i>
• Monitored daily production operations, equipment availability, and plant performance to support efficient, uninterrupted production. • Collected and analysed operational data to support process monitoring, productivity assessment, equipment performance evaluation, and engineering decision-making.	

- Coordinated with cross-functional teams, prepared technical reports, monitored equipment performance, and supported troubleshooting, safety procedures, SOP compliance, and continuous process improvement.

Projects

Comparative Analysis of Hydrogen Production Pathways Using Aspen Plus

Jul 2026 – Present

M.Tech Research Project – Fuel & Energy Systems

IIT (ISM) Dhanbad, Jharkhand

- Developing a comparative Aspen Plus framework for hydrogen production through biomass gasification without WGS, WGS-integrated gasification, and biogas reforming under consistent process assumptions.
- Modelling biomass drying, feed decomposition, gasification, water-gas shift reaction, reforming, gas cleaning, and hydrogen separation to evaluate pathway-level process performance.
- Analysing syngas composition, hydrogen yield, H_2/CO ratio, CO conversion, mass and energy balances, energy efficiency, and exergy efficiency across the considered pathways.
- Performing sensitivity analysis of gasification/reforming temperature, steam-to-feed ratio, and WGS operating conditions to identify favourable operating ranges and compare process performance.

Technical Skills

Fuel & Energy Engineering: Fuel Technology, Energy Conversion, Thermochemical Conversion, Biomass Conversion, Biofuels, Renewable Energy, Energy Systems, Energy Efficiency

Process Engineering & Simulation: Aspen Plus, Process Simulation, Process Modelling, Process Flow Diagrams (PFD), Mass & Energy Balances, Process Evaluation, Sensitivity Analysis, Process Optimisation

Biomass & Hydrogen Processes: Biomass Gasification, Biogas Reforming, Water-Gas Shift (WGS), Syngas Analysis, Hydrogen Production

Engineering Analysis: Energy Analysis, Exergy Analysis, Engineering Calculations, Data Analysis, Technical Reporting

Programming & Software: Python, MATLAB, Microsoft Excel, AutoCAD

Achievements

- **GATE 2025 Qualified** with a score of 356 and All India Rank (**AIR 1133**).
- Secured **4th Merit Rank** in B.Tech Mining Engineering at Rajasthan Technical University (RTU), Kota.